# EE 439: Introduction to Machine Learning
## Project Report: Data First, Models Second

Member 1 Name (ID) & Member 2 Name (ID)

Spring 2026

**Selected Dataset:** [Insert Dataset Name Here]

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**Note: Comment out the later sections that are not relevant for the phase that you are preparing your document for at the moment. For example, the sections for Phase 1, 2 and 3 are commented here. The final report will have all the sections uncommented.**

# 1 Phase 0: Exploratory Data Analysis

*Deadline: April 22*

## 1.1 Data Partitioning [Member Name]

Confirm here that the data was split into a Development Set (80%) and Test Set (20%), and that the Test Set remained untouched during this phase. Give the details on the strategy used to split the dataset.

## 1.2 Visual Exploration and Insights

Include at least 5 informative figures which you think are worth keeping for the analysis and add at least 5 figures which you worked on but didn't consider worth the analysis and give the reasons why you think like that.

### 1.2.1 Illustrative Figure 1 [Member 1 Name]

[Add figure.] Use the following syntax for adding figures:

- Why is this informative?
  *Explain here.*

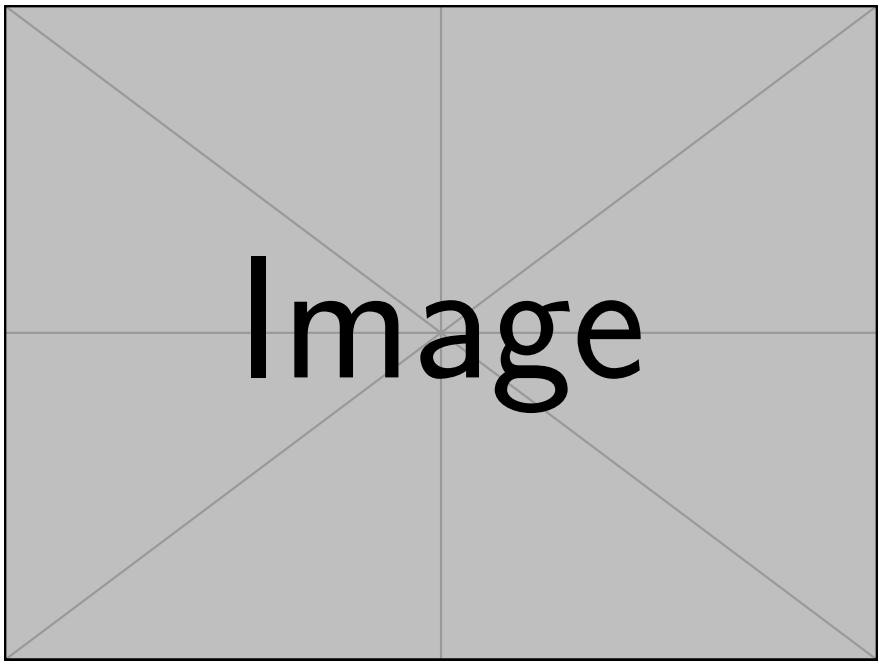
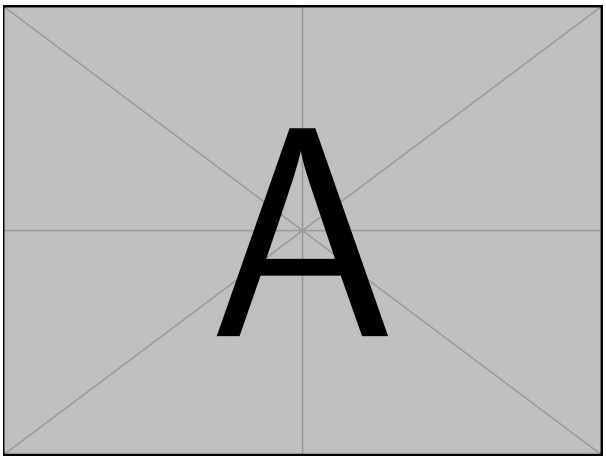
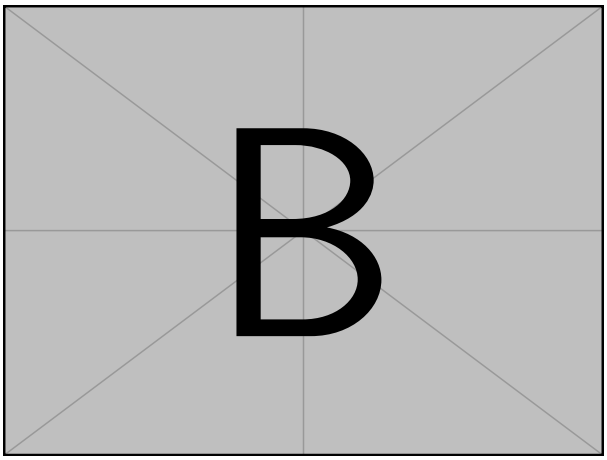



Figure 1: Description of what is shown. (e.g., "The right-skewed target suggests a log-transformation might be needed")?




(a) Feature A vs Target




(b) Feature B vs Target

Figure 2: Comparative analysis of linear vs nonlinear trends.

- What does it inform?
  *Explain here.*
- How does it shape your thought process?
  *Explain here.*

#### 1.2.2 Illustrative Figure 2 [Member 1 Name]

[Add figure.]

- Why is this informative?
  *Explain here.*
- What does it inform?
  *Explain here.*
- How does it shape your thought process?
  *Explain here.*

#### 1.2.3 Illustrative Figure 3 [Member 1 Name]

[Add figure.]

- Why is this informative?
  *Explain here.*
- What does it inform?
  *Explain here.*
- How does it shape your thought process?
  *Explain here.*

#### 1.2.4 Illustrative Figure 4 [Member 2 Name]

[Add figure.]

- Why is this informative?
  *Explain here.*
- What does it inform?
  *Explain here.*
- How does it shape your thought process?
  *Explain here.*

#### 1.2.5 Illustrative Figure 5 [Member 2 Name]

[Add figure.]

- Why is this informative?
  *Explain here.*
- What does it inform?
  *Explain here.*
- How does it shape your thought process?
  *Explain here.*

#### 1.2.6 Non-Illustrative Figure 1 [Member 2 Name]

[Add figure.]

- Why is this non-informative?
  *Explain here.*
- How does it shape your thought process?
  *Explain here.*

#### 1.2.7 Non-Illustrative Figure 2 [Member 2 Name]

[Add figure.]

- Why is this non-informative?
  *Explain here.*
- How does it shape your thought process?
  *Explain here.*

#### 1.2.8 Non-Illustrative Figure 3 [Member 2 Name]

[Add figure.]

- Why is this non-informative?
  *Explain here.*
- How does it shape your thought process?
  *Explain here.*

#### 1.2.9 Non-Illustrative Figure 4 [Member 1 Name]

[Add figure.]

- Why is this non-informative?
  *Explain here.*
- How does it shape your thought process?
  *Explain here.*

#### 1.2.10 Non-Illustrative Figure 5 [Member 1 Name]

[Add figure.]

- Why is this non-informative?
  *Explain here.*
- How does it shape your thought process?
  *Explain here.*

## 2 Phase 1: Hypothesis and Model Justification

*Deadline: May 04 (2–3 pages)*

### 2.1 Structural Observations

Based on Phase 0, describe the problem structure (e.g., linearity, noise levels, spatial clusters). *Add subsections as per your convenience. Make sure to include the member name in the subsection title.*

### 2.2 Hypothesis on Model Complexity

State your hypothesis regarding how much complexity is required to solve this problem without overfitting. *Add subsections as per your convenience. Make sure to include the member name in the subsection title.*

### 2.3 Model Selection

- **Baseline Model [Member 1 Name]:** [e.g., Linear Regression]
- **Main Model [Member 2 Name]:** [e.g., Random Forest]
- **Flexible Comparison [Member 1 Name]:** [e.g., Gradient Boosting]

### 2.4 Evidence Table

*Add rows to the table as per your convenience.*

| Evidence from Phase 0 | Implication for Model Choice | Member Name |
|---|---|---|
| Observation 1... | Implication 1... continued to next line | Name |
| Observation 2... | Implication 2... | Name |

Table 1: Mapping data characteristics to model selection.

## 3 Phase 2: Model Implementation and Evaluation

*Deadline: May 11*

### 3.1 Model Implementation

Give the details about the preprocessing (if any), hyperparameters and the Python library used to run the model.

#### 3.1.1 Baseline Model [Member 2 Name]:

- Preprocessing (*Optional*)
  *Your response.*
- Hyperparameters
  *Your response.*
- Python library
  *Name of the library. e.g., scikit-learn.*

#### 3.1.2 Main Model [Member 1 Name]

- Preprocessing (*Optional*)
  *Your response.*
- Hyperparameters
  *Your answer.*
- Python library
  *Name of the library. e.g., scikit-learn.*

#### 3.1.3 Flexible Comparison [Member 2 Name]:

- Preprocessing (*Optional*)
  *Your answer.*
- Hyperparameters
  *Your answer.*
- Python library
  *Your answer.*

### 3.2 Performance Metrics

Report the metrics (RMSE, Accuracy, F1, etc.) for all three models using cross-validation. *Add subsections as per your convenience. Make sure to include the member name in the subsection title.*

### 3.3 Hypothesis Testing

- Did the results support your Phase 1 hypothesis? [Member 1 Name]
  *Your answer.*
- Compare the performance of the Baseline vs. Main vs. Flexible models [Member 2 Name]
  *Add the comparsion here. Try adding some figures to support your case.*

## 4 Phase 3: Reflection

*Deadline: May 18*

1. Was your original hypothesis correct? [Member 2 Name]
   *Your answer.*
2. Which observations were most predictive of success? [Member 2 Name]
   *Your answer.*
3. Which observations were misleading? [Member 1 Name]
   *Your answer.*
4. Future approach: How would you pivot if you started over? [Member 1 Name]
   *Your answer.*

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## Statement of Contribution

- **Member 1:** (List down the contributions. It should be easy to add them here since all the subsections clearly define the assignment to exactly one member.).

- **Member 2:** (List down the contributions. It should be easy to add them here since all the subsections clearly define the assignment to exactly one member.).